

Machine Learning

Tutorial 1

Exercise 1:

1. What is predictive data analytics?
2. What is supervised machine learning?
3. Machine learning is often referred to as an ill-posed problem. What does this mean?
4. The following table (Table 1) lists a dataset from the credit scoring domain. Underneath the table we list two prediction models consistent with this dataset, Model 1 and Model 2.

Table 1: Loan-Salary Data

ID	Occupation	Age	Ratio	Outcome
1	industrial	39	3.40	default
2	industrial	22	4.02	default
3	professional	30	2.7	repay
4	professional	27	3.32	default
5	professional	40	2.04	repay
6	professional	50	6.95	default
7	industrial	27	3.00	repay
8	industrial	33	2.60	repay
9	industrial	30	4.5	default
10	professional	45	2.78	repay

- (a) Which of these two models do you think will generalize better to instances not contained in the dataset?
- (b) Propose an inductive bias that would enable a machine learning algorithm to make the same preference choice that you made in Part (a).
- (c) Do you think that the model that you rejected in Part (a) of this question is overfitting or underfitting the data?

Algorithm 1 Model 1

```
1: if LOAN-SALARY RATIO > 3.00 then
2:   OUTCOME = default
3: else
4:   OUTCOME = repay
5: end if
```

Algorithm 2 Model 2

```
1: if AGE = 50 then
2:   OUTCOME = default
3: else if AGE = 39 then
4:   OUTCOME = default
5: else if AGE = 30 and OCCUPATION = industrial then
6:   OUTCOME = default
7: else if AGE = 27 and OCCUPATION = professional then
8:   OUTCOME = default
9: else
10:  OUTCOME = repay
11: end if
```

Exercise 2:

1. What is meant by the term inductive bias?
2. How do machine learning algorithms deal with the fact that machine learning is an ill-posed problem?
3. What can go wrong when an inappropriate inductive bias is used?
4. It is often said that 80% of the work done on predictive data analytics projects is done in the Business Understanding, Data Understanding, and Data Preparation phases of CRISP-DM, and just 20% is spent on the Modeling, Evaluation, and Deployment phases. Why do you think this would be the case?

- (a) How many possible models exist for the scenario described by the features in this dataset?
- (b) How many of these potential models would be consistent with this sample of data?

Table 2: Stroke Risk Factors Dataset

ID	High Blood Pressure	Smoker	Diabetes	Heart Disease	Stroke Risk
1	true	false	true	true	high
2	true	true	true	true	high
3	true	false	false	true	medium
4	false	false	false	false	low
5	true	true	true	false	high

- 2- You are using U.S. census data to build a prediction model. On inspecting the data you notice that the RACE feature has a higher proportion of the category White than you expected. Why do you think this might be?
- 3- Why might a prediction model that has very high accuracy on a dataset not generalize well after it is deployed?